

JIAYU (RYAN) ZUO

Sophomore Undergraduate
College of Engineering · EE/BME



HomePage



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Shenzhen, China



Jiayu Zuo



SELF-INTRODUCTION

Hi! I'm a sophomore undergraduate student in the College of Engineering at Southern University of Science and Technology (SUSTech), intending to major in EE/BME. Strong interests in neuroscience and BCI, with goals to explore the mysteries of the brain and develop innovative brain-computer interface technologies.

Currently conducting research at NCC Lab under the supervision of Associate Professor Quanying Liu. For my latest information, please see my personal homepage.

SKILLS

- Programming:** Python, C, Java, MATLAB, LaTeX
- Frameworks:** PyTorch, NumPy, MNE
- Tools:** Git, Docker, Jupyter
- Languages:** Chinese (Native), English (Fluent, CET-4)

RESEARCH PROJECTS (LEADING)

- 05/2026 – Present **EEG-Based Visual Decoding in Real-World Natural Scenes** **NCC Lab, SUSTech**
- Designed a decoding framework targeting single-trial EEG responses to complex natural scene stimuli, eliminating reliance on trial-averaging that limits real-world applicability.
 - Proposed a subject-agnostic alignment strategy to achieve robust cross-subject generalization without per-subject fine-tuning.
- Single-Trial Decoding / Cross-Subject Generalization / Natural Scene / EEG-to-Image
- 03/2026 – Present **Adolescent Brain Development Assessment via Large-Scale fMRI Foundation Model** **NCC Lab, SUSTech**
Core Member
- Building a foundation model pre-trained on a large-scale private fMRI dataset of Chinese school-age children and adolescents to capture developmental brain patterns.
 - Designing downstream evaluation tasks for cognitive and neurodevelopmental assessment, including age prediction and developmental trajectory modeling.
- fMRI Foundation Model / Brain Development
- 09/2025 – Present **EEG-to-Text Decoding via Foundation Model Alignment** **NCC Lab, SUSTech**
- Developed a brain-to-text decoding pipeline aligning EEG representations with large language model embedding spaces via contrastive and generative objectives.
 - Extended decoding to multilingual settings, bridging neural EEG signals and natural language text across languages.
 - Funded by the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds).
- EEG-to-Text / Contrastive Learning
- 06/2025 – 09/2025 **Multi-View 3D Reconstruction System for EEG Electrode Localization** **NCC Lab, SUSTech**
- Developed a precise electrode positioning system using multi-view 3D reconstruction techniques
 - Implemented source localization algorithms and VGGT for visual reconstruction
 - Applied YOLO-based object detection for automated electrode identification
 - Filed one patent application (under review)
- 3D Gaussian Splatting / YOLO / Source Localization

EDUCATION

- 08/2024 - 06/2028 **Bachelor of Engineering** **Southern University of Science and Technology (SUSTech)**
Intended Major: EE/BME
Completed: Calculus, Linear Algebra, College Physics, Fundamentals of Electronic Circuits
In Progress: Data Structures and Algorithms, Probability & Statistics, Signals and Systems

PUBLICATIONS & UNDER REVIEW

- *Equal Contribution*
- BHA/DHA** *(title abbreviated to avoid blind review issues)*
- D. Li*, **J. Zuo***, K. Xie, W. Yang, Y. Liu, C. Wei, & Q. Liu *Under Review (NeurIPS 2026)*
- FlatCLIP** *(title abbreviated to avoid blind review issues)*
- M. Wang, W. Ye, Z. Ning, **J. Zuo**, J. Xia, H. Wen, & Q. Liu *Under Review (NeurIPS 2026)*

HONORS & AWARDS

05/2026	College Students' Innovative Entrepreneurial Training Plan Program	SUSTech
	- Project: <i>Opti-Rhythm: A Digital Sleep Modulation System Based on a Physiological-Topology Brain-Body Coupling Foundation Model</i>. - Selected with a grant of CNY 20,000.	
12/2025	Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds)	SUSTech
	- Project: <i>Foundation Model-Aligned Chinese Semantic Decoding from Brain Signals</i>. - Selected as a top-tier undergraduate project (Top 30%) with a grant of CN¥20,000.	
11/2025	Award of Excellence (Poster Presentation)	Department of Biomedical Engineering, SUSTech
	- Title: <i>MindPilot: EEG-guided Brain Modulation using Closed-loop Visual Stimulation Optimization</i>. - Awarded at the 6th SUSTech BME Research Day.	

COMPETITIONS

05/2026	Finalist (About Top 1%), 2026 Mathematical Contest in Modeling (MCM)	COMAP
	Team Leader & Advisor. Topic: "<i>Artemis's Choice: Chariot or Elevator?</i>".	
12/2025	Winning Prize, 2025(6th) "Greater Bay Area Cup" Financial Mathematical Modeling Contest	GDSTA
	Team Leader. Topic: "<i>Comprehensive Evaluation and Development Analysis of Stablecoins</i>". - GDSTA: Guangdong Association for Science and Technology - Hosted by Guangdong Society for Industrial and Applied Mathematics & Guangdong-Hong Kong-Macau Center for Applied Mathematics(Sun Yat-sen University).	
05/2025	Honorable Mention (H Award), 2025 Mathematical Contest in Modeling (MCM)	COMAP
	Solo Participant. Topic: "<i>Cybersecurity Analysis Based on the Global Cybersecurity Index (GCI)</i>".	
2023	National Second Prize, China High School Mathematical Contest in Modeling	Beijing Normal University
	Solo Participant. Topic: "<i>Game Theoretic Analysis of Medical Resource Runs During the COVID-19 Pandemic</i>".	

RESEARCH EXPERIENCE

Omni-Intelligence

06/2026 – Present

Research Intern working on human preference-aligned post-training for generative models.

Shenzhen
Supervisors: Dr. Chen Wei & Mr. Dongyang Li

Neural Computing and Control Lab (NCC Lab)

09/2024 – Present

Undergraduate Researcher focusing on BCI, CHI, and NeuroAI.

Department of Biomedical Engineering, SUSTech
Supervisor: Associate Professor Quanying Liu